

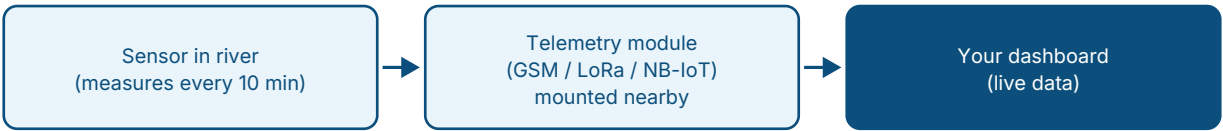
# Water Depth & Temperature Sensors

## Comparison for FunkyForelle deployment

### Sensor Options

| Sensor                   | Measures            | Price (CHF) | Real-time?           | Notes                                                        |
|--------------------------|---------------------|-------------|----------------------|--------------------------------------------------------------|
| HOBO U20L (Onset)        | Depth + temp        | 200–350     | No — logs internally | Very popular in research; manual download required           |
| Solinst Levellogger      | Depth + temp        | 400–700     | No — logs internally | Swiss/European hydrology standard; used by cantonal agencies |
| In-Situ Rugged TROLL 100 | Depth + temp        | 500–900     | With add-on          | Rugged; real-time capable with telemetry module              |
| Keller Series 36W        | Depth + temp        | 300–600     | With add-on          | Swiss manufacturer (Winterthur); likely used by BAFU         |
| Vaisala HydroMet         | Depth + temp + more | 1,000+      | Yes — included       | Higher end; telemetry built in                               |

### What is telemetry?



**Without telemetry:** The sensor stores readings internally. You visit the river physically — weekly or monthly — plug in a cable, and download the data. Cheap to buy, but no real-time warning is possible.

**With telemetry:** A small transmitter box mounted near the sensor reads its output and sends the data wirelessly every few minutes. You see measurements live on a computer. This is what FunkyForelle needs to issue 14-day warnings automatically. One telemetry module is needed per deployment site.

### What CHF 10,000 buys

|                                                                                                                                                                                                                 |                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Without real-time telemetry</b> <ul style="list-style-type: none"><li>• 15–20 basic loggers (HOBO / Solinst)</li><li>• Manual data collection required</li><li>• Good for pilot / validation phase</li></ul> | <b>With real-time telemetry</b> <ul style="list-style-type: none"><li>• 5–8 sensors + telemetry modules</li><li>• Live data stream to FunkyForelle</li><li>• Ready for production deployment</li></ul> |
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Prices are indicative estimates (September 2026). Keller AG (Swiss, Winterthur) is worth approaching for direct pricing and possible BAFU compatibility.